

Errors & Data Formats

Pre-Assessment 3 · 8 questions · 30 points

Name: _____

Date: _____

Score: / 30

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **40 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Which exception?

4 points

Each line crashes. Write the exception name: **ValueError**, **ZeroDivisionError**, **KeyError**, or **FileNotFoundError**. Each one is used once.

Code	Exception
<code>int("ten")</code>	
<code>5 / 0</code>	
<code>{"a": 1}["b"]</code>	
<code>open("missing.txt")</code>	

2. Read the traceback

3 points

TRACEBACK

```
Traceback (most recent call last):
  File "safe_divider.py", line 7, in <module>
    result = divide(10, 0)
  File "safe_divider.py", line 3, in divide
    return a / b
ZeroDivisionError: division by zero
```

(a) What type of error happened?

(1 pt)

(b) On which line number did the error actually happen?

(1 pt)

(c) On which line number was the function called?

(1 pt)

3. Safe Divider

4 points

Write exactly what this program prints.

```
for text in ["8", "zero", "0"]:
    try:
        n = int(text)
        print(24 // n)
    except ValueError:
        print("Not a number")
    except ZeroDivisionError:
        print("Can't divide by zero")
    else:
        print("Success")
    finally:
        print("Next")
```

4. Wrong net

2 points

```
try:
    age = int(input("Age: "))
except ZeroDivisionError:
    print("Numbers only!")
```

(a) The user types `twelve`. What happens?

(1 pt)

(b) Fix the program so it prints `Numbers only!` instead.

(1 pt)

5. Bulletproof input

4 points

```
def ask_number(prompt):
    while True:
        answer = input(prompt)
        try:
            return _____(answer)
        except _____:
            print("Please type a whole number.")
```

(a) Fill in the two blanks.

(2 pts)

(b) The user types `seven`, then `7`. What does the function print, and what does it return?

(2 pts)

6. Pet Registry

4 points

FILE: PETS.CSV

```
name,species,age
Rex,dog,3
Mia,cat,5
Tux,bird,1
```

```
import csv
with open("pets.csv") as f:
    rows = list(csv.reader(f))
print(len(rows))
print(rows[1])

with open("pets.csv") as f:
    for row in csv.DictReader(f):
        if int(row["age"]) > 2:
            print(row["name"])
```

(a) Write exactly what the program prints.

(3 pts)

(b) Why does the program need `int(row["age"])`?

(1 pt)

7. Game Settings

4 points

```
import json
settings = {"sound": True, "level": 3, "players": ["Ana", "Ben"]}
with open("settings.json", "w") as f:
    json.dump(settings, f)

with open("settings.json") as f:
    loaded = json.load(f)
print(loaded["players"][1])
print(type(loaded))
```

(a) What does the first `print` show? (1 pt)

(b) What does the second `print` show? (1 pt)

(c) Inside the file `settings.json`, how is Python's `True` written? (1 pt)

(d) A Python **list** is saved in JSON as a(n) _____. (1 pt)

8. Load the settings

5 points

Write a function `load_settings(filename)` that loads a dictionary from a JSON file and **returns** it. If the file does not exist, it should print `No settings found` and return `{"sound": True, "level": 1}` instead of crashing.

END OF ASSESSMENT